

Gated Recurrent Unit

Contents

- ✓ Introduction to GRU
- ✓ Architecture of GRU
- ✓ Gates in GRU
- ✓ Applications of GRU
- ✓ LSTM Vs GRU

Motivation for Gated Recurrent Unit

RNN



LSTM



Gated
Recurrent Unit
(GRU)

- ✓ Essential for sequential data like text, speech, time series analysis
- ✓ Can't capture long term dependency due to vanishing gradient problem
- ✓ Solved the problem of capturing long term dependencies using complex structure with three gates, forget gate, input gate and output gate.
- ✓ It increased computational cost
- ✓ To overcome this **Gated Recurrent Unit (GRU)** where introduced.
- ✓ Uses LSTM architecture by merging its gating mechanisms

Gated Recurrent Unit

Example:

“My grandparents live in **Germany**, and I visit them every summer, which is why I understand **German** quite well but I am well versed in English as well”

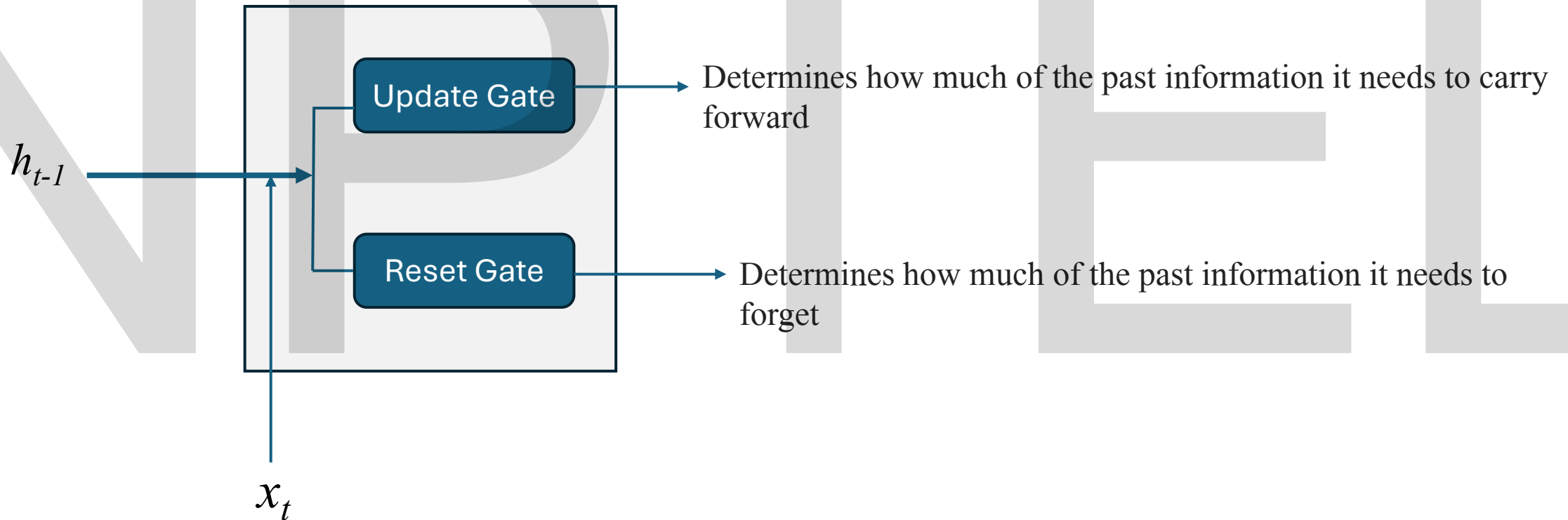
Like LSTM, GRU is designed to model sequential data by allowing information to be selectively remembered or forgotten over time.

Gated Recurrent Units (GRUs) are introduced by Cho et al. in 2014 in the paper Learning Phrase Representations using RNN Encoder-Decoder for Statistical Machine Translation".

Architecture of GRU

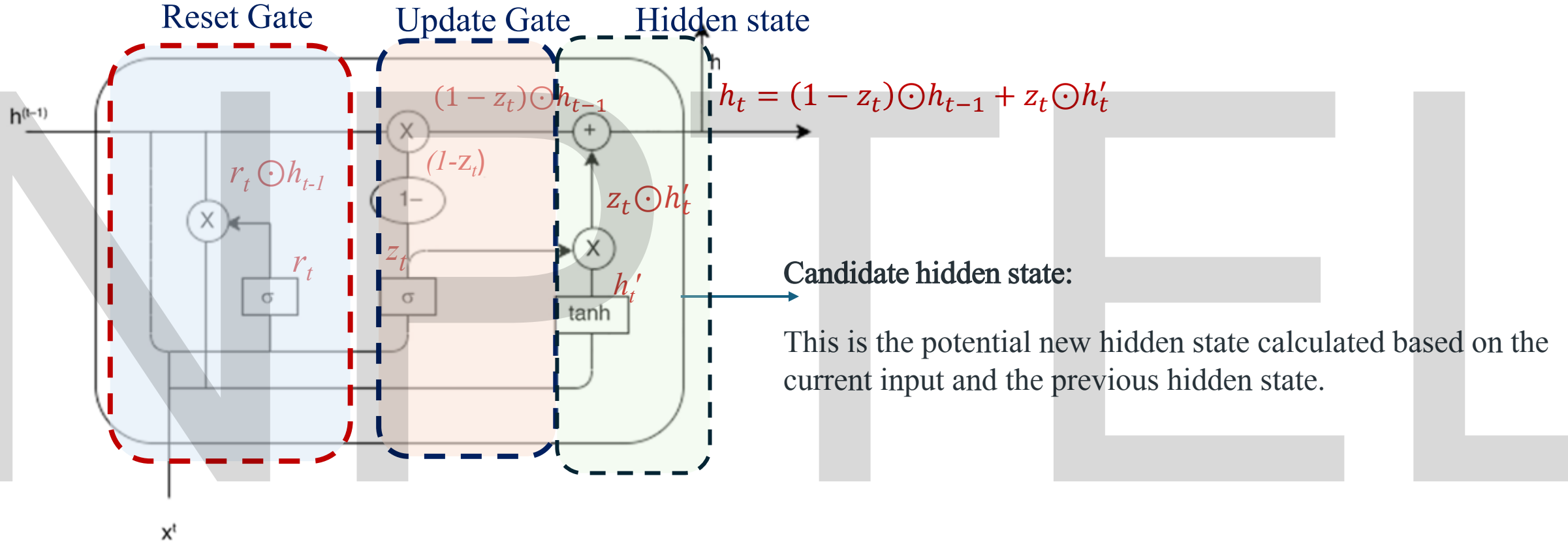
A GRU has two gates

- ✓ Update Gate z_t
- ✓ Reset Gate r_t



Unlike LSTM, GRU does NOT have a separate cell state, it has only hidden state h_t

Architecture of the Reset and Update Gate



$$r_t = \sigma(W_r \cdot [h_{t-1}, x_t])$$

$$h'_t = \tanh(W_h \cdot [r_t \odot h_{t-1}, x_t])$$

$$z_t = \sigma(W_z \cdot [h_{t-1}, x_t])$$

Comparison between LSTM and GRU

LSTM

GRU

Similarity

- Both are types of recurrent neural network (RNN) layers designed to handle sequential data.
- They address the vanishing gradient problem in traditional RNNs by introducing gating mechanisms that allow them to capture long-term dependencies more effectively.

Differences

- | | |
|-----------------------------|-----------------------------|
| ➤ 3 gates are used | ➤ 2 gates are used |
| ➤ More complex architecture | ➤ Architecture simpler |
| ➤ More number of parameters | ➤ No of parameters are less |
| ➤ Training time more | ➤ Training time less |

Where to use LSTM and where GRU?

LSTM

- ✓ Ideal for time-series forecasting, text generation and tasks requiring long-term dependencies

GRU

- ✓ Preferred for cases where computational efficiency is important

Limitation of LSTMs and GRUs

- ✓ Though the problem of long term dependency has been reduced by the use of LSTMs and GRUs but it still exists.

Attention Mechanism

NPTEL

Summary

Next Session

LSTM- Encoder Decoder structure

NIPTELL

Thank you